Table 2. Classification accuracy (%) on adversarial examples. Ensemble models consist of three Resnet-20. For JSMA, the perturbation $\epsilon = 0.2$ on MNIST, and $\epsilon = 0.1$ on CIFAR-10. For EAD, the factor of L_1-norm $\beta = 0.01$ on both datasets.

	MNIST				CIFAR-10			
Attacks	Para.	Baseline	$\text{ADP}_{2,0}$	$\text{ADP}_{2,0.5}$	Para.	Baseline	$\text{ADP}_{2,0}$	$\text{ADP}_{2,0.5}$
FGSM	$\epsilon = 0.1$	78.3	95.5	**96.3**	$\epsilon = 0.02$	36.5	57.4	**61.7**
	$\epsilon = 0.2$	21.5	50.6	**52.8**	$\epsilon = 0.04$	19.4	41.9	**46.2**
BIM	$\epsilon = 0.1$	52.3	86.4	**88.5**	$\epsilon = 0.01$	18.5	44.0	**46.6**
	$\epsilon = 0.15$	12.2	69.5	**73.6**	$\epsilon = 0.02$	6.1	28.2	**31.0**
PGD	$\epsilon = 0.1$	50.7	73.4	**82.8**	$\epsilon = 0.01$	23.4	43.2	**48.4**
	$\epsilon = 0.15$	6.3	36.2	**41.0**	$\epsilon = 0.02$	6.6	26.8	**30.4**
MIM	$\epsilon = 0.1$	58.3	89.7	**92.0**	$\epsilon = 0.01$	23.8	49.6	**52.1**
	$\epsilon = 0.15$	16.1	73.3	**77.5**	$\epsilon = 0.02$	7.4	32.3	**35.9**
JSMA	$\gamma = 0.3$	84.0	88.0	**95.0**	$\gamma = 0.05$	29.5	33.0	**43.5**
	$\gamma = 0.6$	74.0	85.0	**91.0**	$\gamma = 0.1$	27.5	32.0	**37.0**
C&W	$c = 0.1$	91.6	95.9	**97.3**	$c = 0.001$	71.3	76.3	**80.6**
	$c = 1.0$	30.6	75.0	**78.1**	$c = 0.01$	45.2	50.3	**54.9**
	$c = 10.0$	5.9	20.2	**23.8**	$c = 0.1$	18.8	19.2	**25.6**
EAD	$c = 5.0$	29.8	91.3	**93.4**	$c = 1.0$	17.5	64.5	**67.3**
	$c = 10.0$	7.3	87.4	**89.5**	$c = 5.0$	2.4	23.4	**29.6**

Table 3. Classification accuracy (%) on adversarial examples. Ensemble models consist of three Resnet-20.

	CIFAR-100		
Attacks	Para.	Baseline	$\text{ADP}_{2,0.5}$
BIM	$\epsilon = 0.005$	21.6	**26.1**
	$\epsilon = 0.01$	10.1	**14.8**
PGD	$\epsilon = 0.005$	26.6	**32.1**
	$\epsilon = 0.01$	11.7	**18.3**
MIM	$\epsilon = 0.005$	24.2	**29.4**
	$\epsilon = 0.01$	11.2	**17.1**

steps are set to be 10 for BIM, PGD, and MIM, with the step size equals to $\epsilon/10$. The iteration steps are set to be $1,000$ for C&W and EAD, with the learning rate of 0.01. We separately test the ensemble models trained by the baseline, $\text{ADP}_{2,0}$ and $\text{ADP}_{2,0.5}$ methods. As expected, the results verify that our method significantly improves adversarial robustness. However, the improvements are more prominent on MNIST and CIFAR-10 than those on CIFAR-100. This can be explained by Eq. (8): the non-maximal predictions scale with the factor $\frac{1}{L-1}$, which could result in a numerical obstacle in the training phase when L is large. A potential way to solve this problem is to use the temperature scaling method (Guo et al., 2017), as discussed in Appendix B.2. An interesting phenomenon is that when the LED part is inactive as in the $\text{ADP}_{2,0}$ setting, the learned ensemble diversity is still much larger than the baseline (detailed in Appendix B.3). This is because the ensemble entropy part expands the feasible space of the optimal solution, while leaving degrees of freedom on the optimal non-maximal predictions F^k_j, as shown in Theorem 2. Then the ensemble diversity is unlikely to be small in this case. However, the existence of the LED part further explicitly encourages the ensemble diversity, which leads to better robustness.

To show that our method is an orthogonal approach concerning other defenses acting on a single network, we test the compatibility of our method with adversarial training (AdvT), which is the most widely studied defense method (Kurakin et al., 2018b). AdvT augments the training data with adversarial examples in each mini-batch, where the two quite common attacks used to craft these adversarial examples are FGSM (Goodfellow et al., 2015) and PGD (Madry et al., 2018). In Table 4, we demonstrate the classification accuracy of enhanced ensemble models on CIFAR-10. We uniformly sample the perturbation ϵ from the interval $[0.01, 0.05]$ when performing AdvT as in Kurakin et al. (2017b). The ratio of adversarial examples and normal ones in each mini-batch is 1:1, with the batch size of 128. The results demonstrate that our method can further improve robustness for ensemble models, with little extra computation and is universal to different attacks. Besides, when comparing the results in Table 2 and Table 4, we can find that AdvT can also further improve the performance of our method, which means they are indeed complementary.

Table 4. Classification accuracy (%): $\textbf{AdvT}_{\textbf{FGSM}}$ denotes adversarial training (AdvT) on FGSM, $\textbf{AdvT}_{\textbf{PGD}}$ denotes AdvT on PGD. $\epsilon = 0.04$ for FGSM; $\epsilon = 0.02$ for BIM, PGD and MIM.

	CIFAR-10			
Defense Methods	FGSM	BIM	PGD	MIM
$\text{AdvT}_{\text{FGSM}}$	39.3	19.9	24.2	24.5
$\text{AdvT}_{\text{FGSM}} + \text{ADP}_{2,0.5}$	**56.1**	**25.7**	**26.7**	**30.6**
AdvT_{PGD}	43.2	27.8	32.8	32.7
$\text{AdvT}_{\text{PGD}} + \text{ADP}_{2,0.5}$	**52.8**	**34.0**	**36.2**	**38.8**

4.4. Transferability among Individual Models

Due to the transferability of adversarial examples among models (Papernot et al., 2016b), black-box adversaries can craft adversarial examples based on substitute models and then feed these examples to original models to perform the attack. In this sub-section, we show that the ADP training can mitigate the transferability of adversarial examples among the members in the ensemble, as demonstrated in Fig. 3. For each matrix, the (i, j) element is the result of using i-th network as the substitute model to craft adversarial examples and feeding to j-th network as the original model.

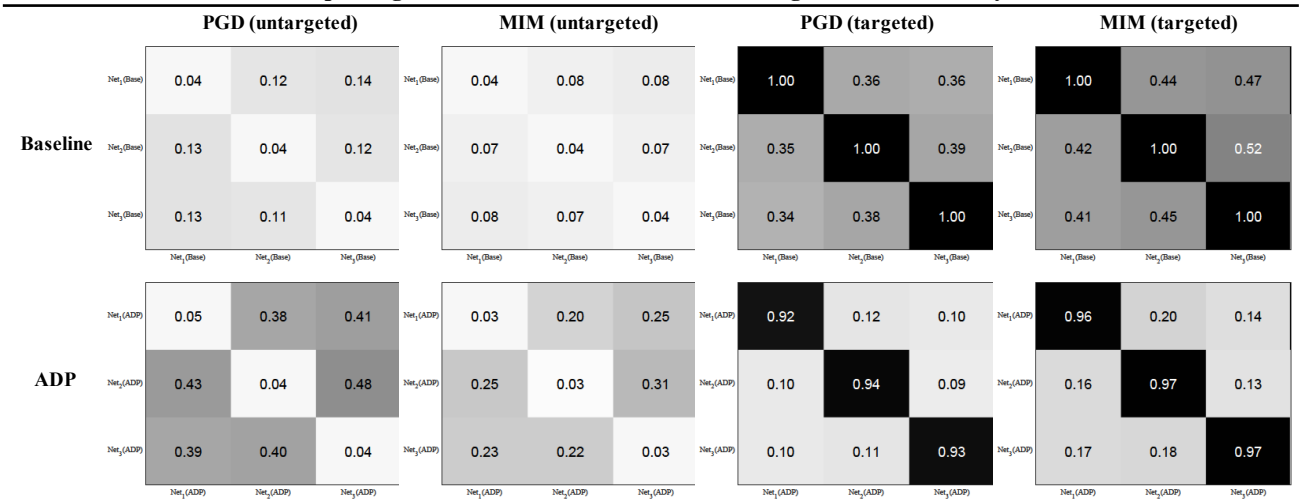
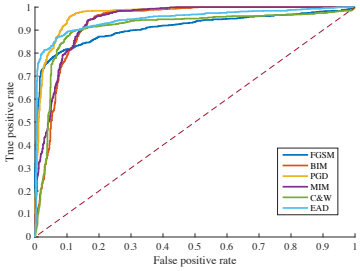


Figure 3. Adversarial transferability among individual models on CIFAR-10. For untargeted attacks, the values are the classification accuracy; For targeted attacks, those are the success rate of adversarial examples fooling classifiers to predict target classes.



Attacks	Para.	AUC-scores (10^{-2})
FGSM	$\epsilon = 0.1$	91.19
BIM	$\epsilon = 0.1$	93.14
PGD	$\epsilon = 0.1$	97.03
MIM	$\epsilon = 0.1$	94.09
C&W	$c = 1.0$	90.98
EAD	$c = 20.0$	94.84

Figure 4. The ROC-curves on 1,000 test examples and 1,000 adversarial examples of CIFAR-10. The model is trained by $\text{ADP}_{2,0.5}$.

	CIFAR-10		
Attacks	Para.	Baseline	$\text{ADP}_{2,0.5}$
Normal	-	93.6	**93.8**
FGSM	$\epsilon = 0.02$	42.0	**58.4**
BIM	$\epsilon = 0.01$	31.6	**41.8**
PGD	$\epsilon = 0.01$	37.4	**44.2**
MIM	$\epsilon = 0.01$	37.1	**47.5**
C&W	$c = 0.01$	52.3	**56.5**
EAD	$c = 1.0$	20.4	**65.3**

Table 5. Classification accuracy (%) on adversarial examples. Ensemble models consist of *five* Resnet-20 such that $5 \nmid 9$.

We apply PGD and MIM as the attack methods, which are the two most commonly used attacks in the black-box setting (Kurakin et al., 2018b). The perturbation parameters for both attacks are set to be $\epsilon = 0.05$. For a complete analysis, we test untargeted mode and targeted mode. For untargeted mode, adversaries only aim to make the models predict wrong labels, and the evaluation criterion is the classification accuracy of the models on adversarial examples. For targeted mode, the evaluation criterion becomes the success rate of fooling the models to predict specific target label. As the results indicate, the ADP training can significantly alleviate the transferability among the members of the ensembles in both untargeted and targeted modes, which can further lead to better robustness for the ensemble models.

4.5. Ensemble Diversity as the Detection Metric

When the adversarial perturbation becomes larger, neither existing defenses nor our method can avoid misclassifying on white-box adversarial examples (Athalye et al., 2018). Then another possible solution is detecting and filtering out adversarial examples instead (Pang et al., 2018a). In Fig. 4, we show the ROC-curves and corresponding AUC-scores when applying ensemble diversity as the detection metric to filter out adversarial examples. The models are trained by $\text{ADP}_{2,0.5}$. Here the parameters used for all attacks are much larger than those in Table 2, which can make trained models misclassify with nearly 100% rate. The results show the effectiveness of ensemble diversity as the detection metric on ADP-trained models, which generalizes our method to a more extensible defense in the adversarial setting.

4.6. The Cases of $K \nmid (L-1)$

To further investigate the flexibility of our method, we consider the cases of $K \nmid (L-1)$. We test on CIFAR-10 with $L = 10$ and construct an ensemble consisting of five individual members, i.e., $K = 5$. It is obvious that $5 \nmid 9$. The results are demonstrated in Table 5. We can find that even if the condition $K \mid (L-1)$ in Corollary 1 does not hold, the ADP training can still largely outperform the baseline method, which suggests that our method is practically insensitive to the relationship of K and L, even if L is not large, and thus can be applied in more general cases.

5. Conclusion

In this paper, we propose the ADP training method. Compared to previous efforts that focus on enhancing a single network, we provide an orthogonal approach to further improve a practical ensemble defense, which implies that ensemble diversity could be an important consideration to promote when designing robust ensemble systems.

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